View from Japan

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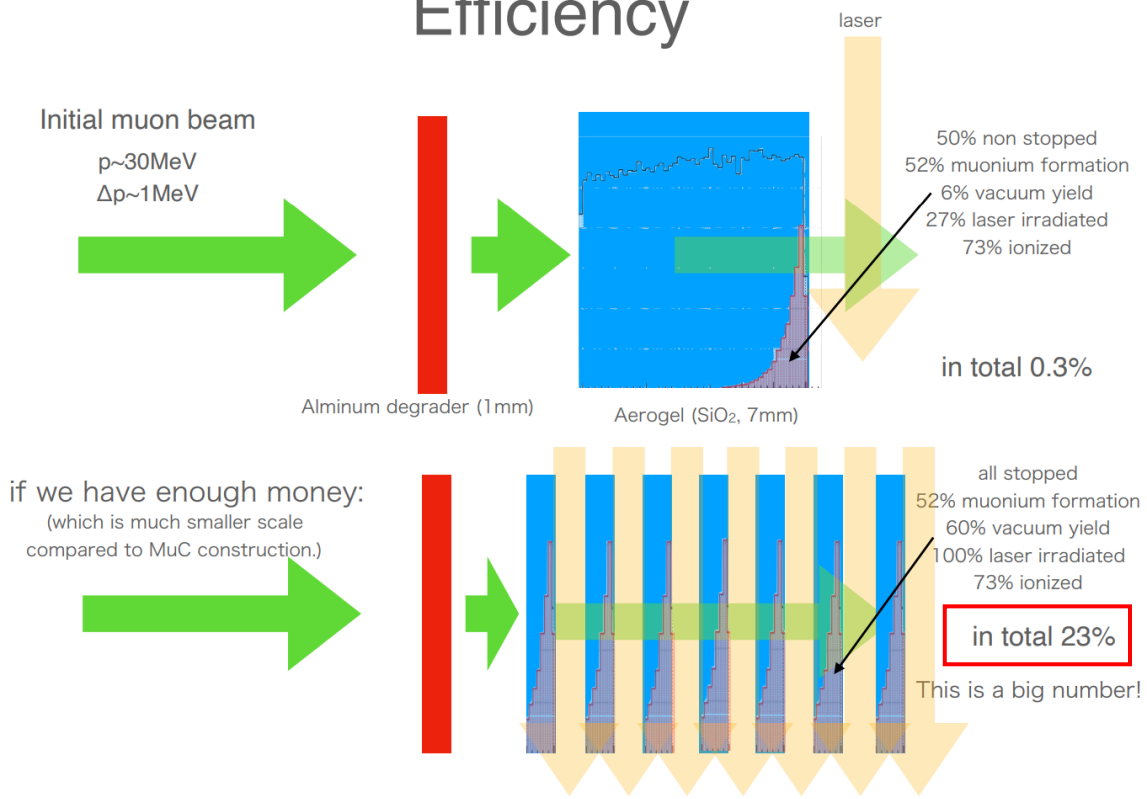
Efficiency



Hamada, Y., Kitano, R., Matsudo, R., Takaura, H., & Yoshida, M. (2022). μTRISTAN. *Progress of Theoretical and Experimental Physics*, *2022*(5), 053B02. https://doi.org/10.1093/ptep/ptac059

muonium atoms are thermally diffused from the aerogel and ionized by the Lyman-α laser. The ionized μ^+ has a very small momentum spread, and is called the ultra-cold muon.

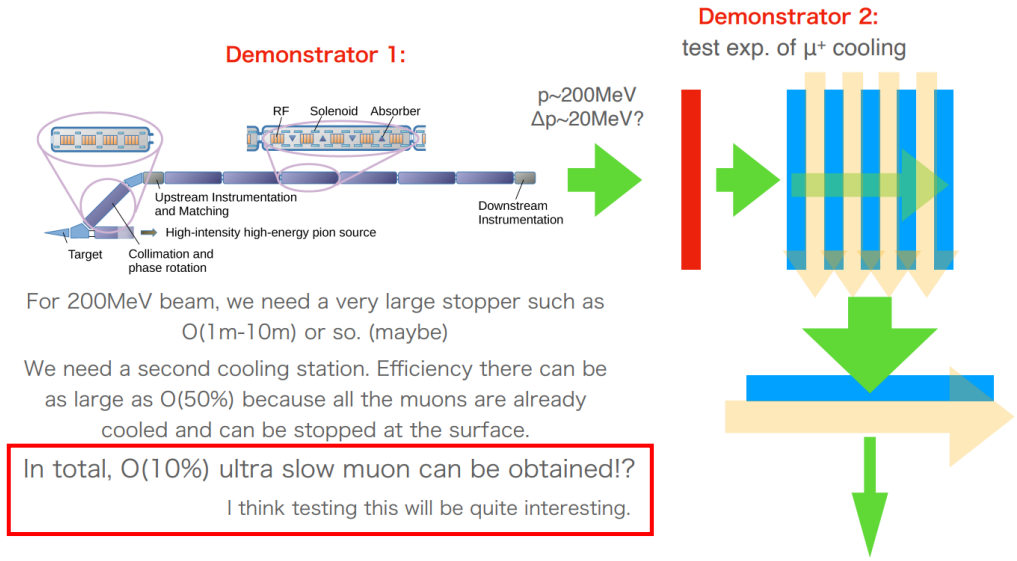
The production efficiency of the ultra-cold muons in total is estimated to be 14% from the following estimated efficiencies: 52% for muonium formation, 60% for the vacuum yield, 60% after the loss of muoniums due to decay, and 73 % for the laser ionization.[2]

The ultra-cold muons spread over 10 m are in turn collected and transported to the second aerogel target with a size of a few centimeters. There, muonium is formed again and re-ionized by laser. The two bunches of muons from the pion production ring are combined at the second muon target by tuning the timing of the laser irradiation. The production efficiency of ultra-cold muons at the second muon target is assumed to be 50%. Such high efficiency should be possible since a very thin target is used for the uniform muon energy. In total, the efficiency in obtaining ultra-cold muons from the stopped pions is 6.8%.

Combining the 50% efficiency of the pion stopping and the pion production rate of 0.016/pro-

Now, why don't we consider this?

Demonstrator[2]



For 200MeV beam, we need a very large stopper such as O(1m-10m) or so. (maybe)

We need a second cooling station. Efficiency there can be as large as O(50%) because all the muons are already cooled and can be stopped at the surface.

In total, O(10%) ultra slow muon can be obtained!?

I think testing this will be quite interesting.